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Influence of the Filling Factor and Size of Coated Spheres on Effective Refractive Index in Nanosphere Dispersed Liquid Crystal Metamaterial

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We study the tunability of the effective refractive index in nanosphere dispersed liquid crystal (NDLC) metamaterial by means of the filling factor and size ratio of coated spheres. We show that small variations of those parameters lead to strong changes in the optical properties (effective refractive index) of NDLC.

Keywords: Nanosphere dispersed liquid crystal, Metamaterials, Effective refractive index

INTRODUCTION

The studied NDLC metamaterial [1, 2] is based on nematic liquid crystal (LC) in which the nanospheres with core – shell structure are randomly distributed (Fig. 1). In this material the effective refractive index depends on the composition and dimensions of the coated spheres, filling factor, and on the angle $\theta(\vec{r})$ which makes the long axis of the nematic liquid crystal (NLC) molecule at point $\vec{r}$ with the direction of probing light. The real part of effective refractive index $n'_{eff}(\theta)$ depends on the wavelength λ of the probing light.

The tunability of optical parameters of NDLC metamaterial is of particular importance in view of potential applications. Recently, we have used Monte

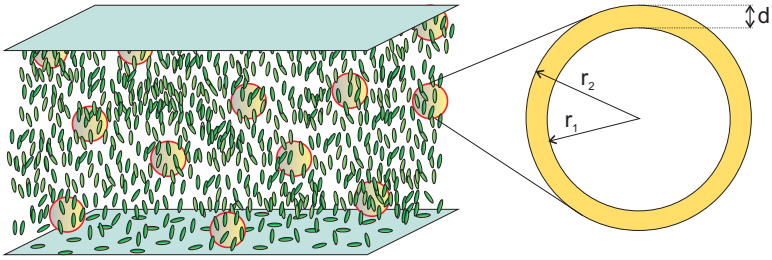

FIGURE 1
Model of the system: liquid crystal and dispersed nanospheres (left). Coated nanosphere (right).

Carlo simulations to study the perspectives for tunability resulting from joint action of electric field and anchoring forces [3, 4].

The aim of this paper is to analyze the dependence of $n'_{eff}(\theta, \lambda)$ on the filling factor and on the size of coated nanospheres.

METHODOLOGY

The effective refractive index of NDLC system was calculated using the method presented in papers [1, 5]. Electromagnetic wave scattering from a single coated sphere was calculated using Mie theory [6] while the effective scattering from an ensemble of coated spheres dispersed in NLC matrix was calculated using Maxwell Garnet mixing rule [6, 7, 8]. This approach relates an effective refractive index of NDLC metamaterial to the parameters of its constitutive components: coated nonmagnetic spheres and host NLC system. The composition of a single coated sphere is presented in Fig. 1. The core with radius r_1 is made from polaritonic material (such as $LiTaO_3$) with relative permittivity

$$\epsilon_1(\omega) = \epsilon_1(\infty)\left(1 + \frac{\omega_L^2 - \omega_T^2}{\omega_T^2 - \omega^2 - i\omega\gamma_1}\right). \tag{1}$$

Here $\epsilon_1(\infty)$ denotes the high-frequency dielectric constant, γ_1 – damping coefficient, ω_T and ω_L – transverse and longitudinal phonon frequencies, respectively. Required relative permittivity is of the order of magnitude 10^2 [5]. Bulk cavity resonance in the core is responsible for negative effective permeability of the assembly of spheres at particular electromagnetic wavelengths. The relative permittivity of the semiconductor shell with outer radius r_2 is described by Drude model:

$$\epsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma_2}, \tag{2}$$

where ω_p denotes plasma frequency and γ_2 – the damping coefficient. The shell is responsible for negative effective permittivity, with ω_p close to frequency for which refractive index attains negative values. Its value is chosen to provide both negative effective permittivity and permeability in some interval of wavelengths [5]. Values of all the material parameters used in this paper can be found in Ref. [1]

Permittivity of the liquid crystal host for extraordinary polarized wave depends on above introduced angle θ between directions of incident light wave vector and director of NLC [2]:

$$\epsilon_3(\theta) = \frac{\epsilon_\parallel \epsilon_\perp}{\epsilon_\parallel \cos^2(\theta) + \epsilon_\perp \sin^2(\theta)}, \tag{3}$$

where $\epsilon_\parallel$ and $\epsilon_\perp$ are the relative permittivities for light polarized respectively parallel and perpendicular to the director axis. In pure NLC host (without spheres) $k_3 = \sqrt{\epsilon_3}\, k_0$.

Electromagnetic wave Mie scattering was calculated in dipolar (electric and magnetic) approximations, with amplitudes of scattered fields proportional to coefficients a_1 and b_1, respectively. The detailed expressions can be found, e.g., in Ref. [6]. Those coefficients enter the formulas for effective permittivity and permeability of NDLC calculated in effective medium theory [1]:

$$\epsilon_{eff} = \epsilon_3 \left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1} \right), \tag{4}$$

$$\mu_{eff} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}. \tag{5}$$

Those formulas are valid only for small number density $N = 3f/4\pi r_2^3$ of spheres (f is the filling factor, see below) and in long-wavelength limit $2\pi/k_3 \gg r_2$. The effective refractive index reads:

$$n_{eff} = \pm\sqrt{\epsilon_{eff}\ \mu_{eff}}, \tag{6}$$

where we have used the root with a positive imaginary part. Khoo et al. have shown [1] that for some values of the parameters of the model there is a range of frequencies ω and angles θ for which:

$$n'_{eff}(\omega, \theta) \equiv \Re(n_{eff}(\omega, \theta)) < 0. \tag{7}$$

EFFECTS DUE TO VARIATION OF THE FILLING FACTOR

This Section is devoted to the study of the influence of the filling factor on $n'_{eff}(\theta)$ for two wavelengths. Let N denote the number density of spheres:

$$N = \frac{m}{V_u}, \tag{8}$$

where m stands for the number of spheres in a unit volume V_u. The filling factor f is defined as:

$$f = \frac{4}{3}\pi r_2^3 N = m\frac{V_s}{V_u}, \tag{9}$$

where V_s denotes the volume of the coated nanosphere.

In previous papers [1, 3] the value $f = 0.12$ was used. Unluckily, it is too high for an experimental implementation. For this reason it is important, having in view potential applications, to study the effective index $n'_{eff}(\theta, \lambda)$ for lower values of f.

Fig. 2 shows the plots of $n'_{eff}(\theta)$ for a few values of the filling factor, calculated for $\lambda = 2820$ nm. For a high value of the filling factor ($f = 0.12$) the effective refractive index is negative in a wide interval of angles θ; the

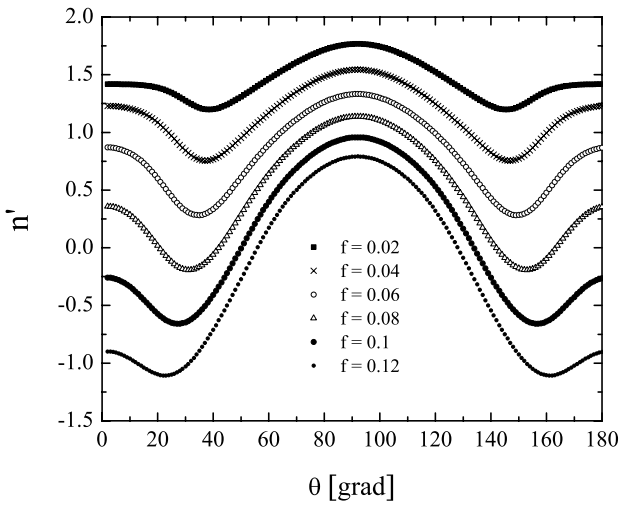

FIGURE 2
Dependence of effective refractive index $n'_{eff}(\theta)$ on angle θ for several values of the filling factor f (from 0.02 to 0.12) for $\lambda = 2820$ nm.

gradients $\partial n'_{eff}/\partial\theta$ have high values [4]. For $f = 0.08$ negative values of n_{eff} appear in narrow intervals of θ. For $f \leq 0.06$ effective refractive index is positive. In the cases $f = 0.04$ and 0.06 refractive index can take values lower than 1, while for $f \leq 0.02$ we find that $n'_{eff} > 1$. The plots of $n'_{eff}(\theta)$ have a minimum which shifts its position from $\theta \approx 20^o$ for $f = 0.12$ to $\theta \approx 35^o$ for $f = 0.02$. At the same time the plots are shifted to higher values of refractive index. The amplitude of the effective refractive index variation, i.e., the difference between its minimal and maximal value, is an increasing function of f.

The shape of the plot of $n'_{eff}(\theta)$ depends strongly on the wavelength λ. Fig. 3 shows the plots of $n'_{eff}(\theta)$ calculated for $\lambda = 2850$ nm. General features of both plots are similar. This time, however, the upward shift of the curves as f decreases is more complicated. Moreover, negative values of n'_{eff} exist only for $f = 0.12$. Refractive index attains values lower than 1 only for $f \geq 0.06$; for lower values of f we find that $n'_{eff} > 1$.

EFFECTS DUE TO VARIATION OF SHELL THICKNESS

The second parameter used for tuning of the effective refractive index is the thickness of the shell:

$$d = r_2 - r_1. \tag{10}$$

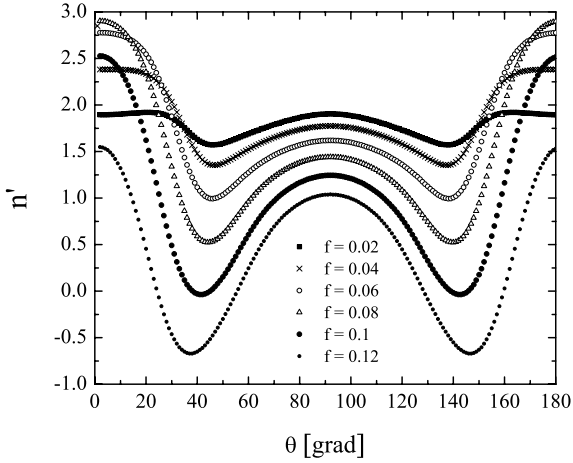

FIGURE 3
Dependence of effective refractive index $n'_{eff}(\theta)$ on angle θ for several values of the filling factor f (from 0.02 to 0.12) for $\lambda = 2850$ nm.

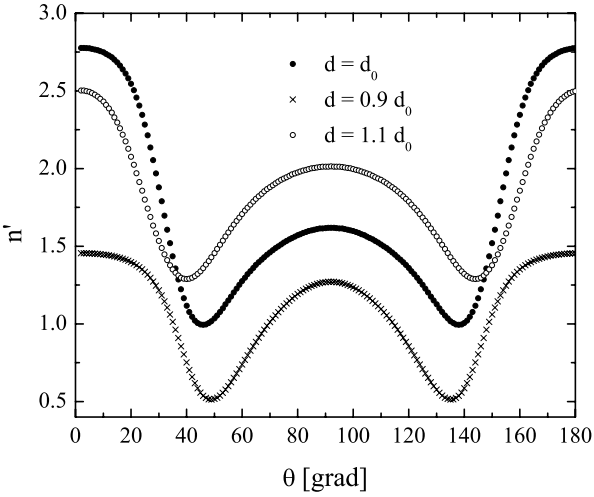

FIGURE 4
Dependence of effective refractive index n'_{eff} on the angle θ for wavelength $\lambda = 2850$ nm and filling factor $f = 0.06$ for thicknesses of the shell equal to $0.9\,d_0$, d_0 and $1.1\,d_0$.

The results of the simulations are compared with the results reported in papers [1, 3] where

$$d_0 = 0.143\ \mu\text{m} - 0.13\ \mu\text{m} = 0.013\ \mu\text{m}. \tag{11}$$

We discuss three cases: $d = 0.9\,d_0, d_0, 1.1\,d_0$. Fig. 4 shows the corresponding plots of $n'_{eff}(\theta)$ calculated for $\lambda = 2850$ nm and for the filling factor $f = 0.06$. Strong dependence on d is observed. For example, a variation of the shell thickness by 10 % changes the value of the refractive index from 0.5 to 1 for chosen angle θ. The general tendency is that an increase of d is accompanied by an increase of refractive index in the central part of the plot, while this relation is more complicated close to $\theta = 0$. Thickness d_0 is specific in that it provides the largest value of the amplitude of the index variation.

CONCLUSIONS

We have shown that the effective refractive index n'_{eff} of NDLC metamaterial is strongly dependent on the filling factor and on the thickness of the shell of nanospheres. Those parameters enrich the set of parameters for an effective tuning of refractive index reported recently in Ref. [3]. They provide the means for a design of specific spatial distributions of refractive index oriented onto

practical applications. In particular, one can curve optical space using the framework of optical transformation [9, 10] and mold the flow of light in many unusual ways like, e.g. in the cases of optical black holes [11] and optical cloaking [12, 13]. The latter is at progress now and will be reported elsewhere.

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